AKTU B.TECH. FIRST YEAR CSE STREAM

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Quantum Physics and Applications

Unit	Description	Weightage
1.	**Fundamentals of Quantum Mechanics:** Inadequacy of classical mechanics, de-Broglie concept of matter waves, Davisson and Germer experiment, wave packet, phase velocity, group velocity, Heisenberg's uncertainty principle and its applications, time-dependent and time-independent Schrodinger wave equations, particle confined in one dimensional box, wave function and its physical interpretation.	20%
2.	**Quantum Theory of Solids:** Free electron theory, Fermi-Dirac distribution, energy bands in solids, band gap, conductors, semiconductors and insulators, effective mass, density of states, intrinsic and extrinsic semiconductors, conductivity of semiconductors, Fermi level.	20%

3.	**Quantum Computing and Communication:** Quantum computation, quantum states, quantum superposition, quantum entanglement, decoherence, quantum bits (qubits), physical realization of qubits (solid state qubits, photonics qubits), basics of quantum gates, single-qubit gate, multi-qubit gate, quantum cryptography, quantum key distribution (QKD).	20%
4.	**Photonics** **Fiber optics:** Fundamental idea about optical fiber, acceptance angle, acceptance cone, numerical aperture, principle of light propagation in optical fiber, attenuation and attenuation coefficient in fibre. **Laser:** Interaction of radiation with matter (absorption, spontaneous and stimulated emission), principle of laser operation, Einstein's coefficients, Ruby laser, He-Ne laser, laser applications.	20%
5.	**Superconductors:** Temperature dependence of resistivity and critical field in superconducting materials, Meissner effect, Type I and Type II superconductors, persistent current, high temperature superconductors, properties, applications of superconductors. **Nano-materials**: Introduction and properties of nano-materials, basic concept of quantum dots, quantum wires and quantum well, applications of nano-materials.	20%

Unit	Description	Weightage (%)
1.	**Chemistry of Hardware Materials for Computing and Electronic Devices:** Introduction to materials used in computers and electronic devices; semiconductors and conductors; silicon and its properties; basic concepts of doping; printed circuit board (PCB) materials; soldering materials; insulating and dielectric materials; reliability and corrosion of electronic components.	20%
2.	**Unit II – Energy Storage and Power Chemistry:** Fundamentals of electrochemistry; primary and secondary batteries; lithium-ion batteries and their applications in laptops, mobiles and EVs; battery safety and charging principles; supercapacitors; fuel cells; introduction to energy harvesting materials and wireless charging technologies.	20%

3.	**Unit III – Sensors and Chemical Detection:** Principles of chemical sensing, electrochemical sensors (amperometric, potentiometric), gas sensors (metal oxide, optical, MEMS-based), biosensors (enzyme, immunosensor, DNA sensor), RFID tag chemistry, chemical detection in environmental monitoring, electronic nose (e-nose) technology	20%
4.	**Unit IV – Nanomaterials and Functional Materials for Smart Systems:** Introduction to nanotechnology; carbon nanotubes (CNTs); graphene; quantum dots; smart coatings; self-cleaning and hydrophobic surfaces; photochromic and thermochromic materials; shape-memory polymers; applications in flexible electronics and smart devices.	20%
5.	**Unit V – Green Chemistry and E-waste Management:** Principles of green chemistry; environmentally friendly electronic materials; biodegradable electronics; e-waste generation and hazards; recycling of electronic waste; recovery of valuable metals; RoHS and WEEE directives; sustainable manufacturing and circular economy in electronics.	20%

Unit	Description	Weightage (%)
1.	**Calculus:** Introduction of Limits, continuity and differentiability for function of two variables, Partial derivatives, Composite functions, Total derivative, Euler's Theorem for homogeneous functions, Change of variables Expansion for function of two variables, Jacobians, properties of Jacobian (without proof), Maxima & Minima for a function of several variables, Lagrange's method of multipliers. Approximation of Errors. **Applications:** AI model optimization, Cloud computing performance, Data processing systems, Machine learning optimization, Neural network tuning, Resource allocation problem.	20%

2.	**Multiple Integrals**: Evaluation of double and triple integrals, change of order of integration, Change of variable (double integral), Beta and Gamma functions, Properties of Beta and Gamma function, Dirichlet,s Integral. Application of double integrals and Triple Integral to find area and volume. **Applications:** Image processing, Data density estimation, AI probability distribution.	20%
3.	**Vector Calculus:** Scalar point function, Vector point function, Gradient of a scalar field, Directional derivatives, Divergence and curl of a vector function and their application to solenoidal and irrotational vectors respectively. Integration of Vectors, Stokes Theorem and Gauss Divergence Theorem (Without Proof). **Applications:** AI heat mapping, Robotics path planning, Computer graphics vector fields.	20%
4.	**Matrices:** Introduction to Matrices, Complex Matrices: Hermitian, Skew Hermitian and Unitary Matrix. Elementary Transformation, Inverse by E-transformation, Rank of a Matrix by Echelon Form, Solution of system of linear Equations by direct method (Gauss Elimination Method), and Iterative method (Gauss Seidel Method). Eigen Values & Eigen Vectors, Similarity Transformation, Diagonalization, Stability Interpretation. **Applications:** Error detection and correction codes, Computer Graphics Rendering , Loop Current Analysis, Principal Component Analysis (PCA), Face recognition systems.	20%
5.	**Introduction to Vector Space:** Definition of Field, Introduction to Vector Spaces, Basic Properties of Vector Spaces, Sub spaces, Linear dependence and independence of vectors, Basis and Dimension. **Application:** Engineering Data Modeling and Compression, Signal representation and processing, Noise reduction using vector space concepts .	20%

COs	Course Outcome Statement	KL
CO1	Apply multivariable calculus for optimization in AI, ML, and cloud computing.	K3

CSE Stream

Unit	Description	Weightage
1.	**Differential Equations:** Linear differential equation of nth order with constant coefficients, Cauchy-Euler equation, solution of second order linear differential equations with variable coefficient by the method of variation of parameters, change of independent variable, normal form. **Applications:** Cloud Computing, Computer Networks, Cyber Security	20%
2.	**Complex Analysis:** Functions of Complex Variables, Limit Continuity and Differentiability, Analytic Function, Cauchy-Riemann Equations (Cartesian and Polar Form), Harmonic Function, Milne-Thomson Method, Complex integrals, Cauchy-	20%

	Integral theorem (without proof), Cauchy integral formula (without proof), Taylor's series and Laurent's series, Singularities, Residue of a function, Cauchy Residue Theorem, Evaluating the Real definite integrals. **Applications:** Image Processing and Computer Design, Quantum Computing	
3.	**Laplace Transform:** Definition of Laplace Transform, Existence Conditions, Linearity Property, Laplace Transform of some standard functions, I and II Shifting theorems (without proof), Change of Scale Property, Laplace transform of derivatives and integrals, Periodic Functions, Unit Step Function, Inverse Laplace Transform, Partial Fraction Method, Convolution theorem (without proof), Application of Laplace transform to solve ordinary differential equation. **Applications:** Database Systems, Algorithmic Modelling	20%
4.	**Fourier Series& Fourier Transform:** Periodic function, even and odd function, Dirichlet's Condition for the Existence of a Fourier Series, Fourier series expansion of a function in the interval *c* to *c+2l*,Change of Interval, Half Range Series: Half range sine and cosine series in the interval *-l* to *l*, Parseval's Identity. Fourier Integral theorem, Fourier transform, Inverse Fourier Transform, Finite Fourier Sin and Cosine transform. **Applications:** Computer Graphics, Cyber Security, Digital Image Processing, Machine Learning Feature Extraction, Image compression.	20%
5.	**Applied Linear Algebra:** Eigen Value Decomposition and Singular Value Decomposition, Matrix factorization, LU Decomposition (Dolittle method), QR Decomposition, Linear transformation, Matrix representation of Linear Transformation, Kernel and Range, Introduction to inner product space, Norm of a vector. **Application:** 2D & 3D graphics, Neural Networks, Cryptography	20%

CSE Stream

Unit	Description	Weightage
1.	**Unit I – Basic Electrical Quantities and DC Circuits:** Electric charge, current, voltage, EMF, resistance; Ohm's law; resistors in series and parallel; voltage divider and current divider; power and energy in a DC circuit; Kirchhoff's Voltage Law (KVL) and Kirchhoff's Current Law (KCL); simple mesh and nodal analysis; Thevenin's theorem (concept and simple problem	20%
2.	**AC Circuits:** Generation of AC voltage; sinusoidal waveform — peak, RMS, and average values; form factor; phasor representation (simple); AC through pure R, pure L, pure C; series RL and RC circuits; impedance; active power (P), reactive power (Q), apparent power (S); power factor (PF); power factor improvement using capacitors; measurement of power in single-phase AC circuit using wattmeter.	20%

3.	**Three-Phase Systems:** Advantages of three-phase supply over single-phase; generation of balanced three-phase voltage; star (Y) and delta (Δ) connections; phase and line voltages and currents; power in balanced three-phase circuits; two-wattmeter method for power measurement; concept of neutral wire	20%
4.	**Magnetic Circuits, Transformers:** Magnetic flux, flux density, MMF, reluctance; simple series magnetic circuit; Faraday's law of electromagnetic induction; self-inductance and mutual inductance; principle of operation of single-phase transformer; EMF equation; ideal transformer — voltage ratio, current ratio, turns ratio; basic equivalent circuit of single phase transformer, losses (core and copper), efficiency	20%
5.	**Electrical Machines and Emerging Electrical Systems:** DC motor — principle, back EMF, torque, speed control concept; Types of DC motors and their application; BLDC motor — basic concept and applications , three-phase induction motor — rotating magnetic field, slip, torque; Application of three phase induction motors, introduction to solar photovoltaic (PV) system- Single diode model, current-voltage characteristics, PV characteristics, electrical safety in laboratory	20%

Unit	Description	Weightage
1.	**Diodes and Power Supply Circuits:** p-n junction — concept of depletion layer, forward and reverse bias; ideal diode vs real diode; V-I characteristics; diode as a switch; half-wave and full-wave bridge rectifier; filter capacitor; Zener diode — Zener breakdown, voltage regulator circuit; LED — working and current-limiting resistor; applications: USB charger rectifier, 5V/3.3V regulated supply for Arduino/Raspberry Pi, LED indicators in electronics products	20%
2.	**Transistors as Switches and Amplifiers:** Bipolar Junction Transistor (BJT) — Construction & VI characteristics of BJT, NPN and PNP, three regions of operation (cut-off, active, saturation); BJT as a switch (digital control of relay, buzzer, motor); BJT as a common-emitter amplifier (concept and gain);	20%

	Introduction to FETs, MOSFET — N-channel MOSFET, gate-source voltage, MOSFET as a switch; why MOSFET is preferred in digital ICs and power electronics; applications: transistor switch for driving a relay in home automation, MOSFET in motor speed control (PWM concept), transistor in class-A amplifier for audio	
3.	**Operational Amplifier (Op-Amp):** Ideal op-amp characteristics — infinite gain, infinite input impedance, zero output impedance; practical characteristics of 741 IC, inverting amplifier — voltage gain formula; non-inverting amplifier — gain formula; voltage follower (buffer); summing amplifier; comparator — op-amp as comparator, threshold detection; applications: sensor signal amplification (temperature, light, pressure), audio amplifier, automatic night lamp comparator circuit, differential amplifier for ECG/EEG front-end (concept)	20%
4.	**Digital Electronics Basics:** Number systems — binary, decimal, hexadecimal (conversion only); logic levels (HIGH/LOW, 0/1); basic logic gates — AND, OR, NOT, NAND, NOR, XOR with truth tables; Boolean algebra — basic laws and De Morgan's theorems; k-map (upto 4 variables), half adder and full adder (truth table and logic diagram); multiplexer concept; flip-flop — SR and D flip-flop (concept only); applications: binary addition in CPU ALU, MUX in data routing, digital signals in microcontroller I/O, basics of PLA / PLD devices.	20%
5.	**Electronic Systems and Emerging Applications:** Microcontroller overview — what is a microcontroller, GPIO pins, ADC- ADC types – digital ramp type & flash type ADC, PWM (concept only, not programming); communication interfaces — UART, SPI, I2C (what they do, not detailed protocol); sensors and signal chain — sensor → conditioning (op-amp) → ADC → microcontroller; wireless communication basics — Bluetooth, Wi-Fi, Zigbee (concept); power management in electronics — LDO regulator, DC-DC converter (concept); applications: IoT weather station, smart home sensor node,	20%

CSE Stream

Unit	Language	Topics	Weightage
1.	C	**Programming Basics (C):** 1. Structure of a C program; compilation and execution 2. Data types, variables, constants, operators (arithmetic, relational, logical) 3. Input/output — scanf/printf in C 4. Type conversion; simple expressions *Real connect: C powers embedded systems, microcontrollers, and OS kernels.*	20%
2.	C	**Decision Making and Control Flow (C):** 1. Control Statements 2. Loops — for, while, do-while; break and continue 3. Functions — definition, call, return; call by value	20%

		4. Recursion — factorial and Fibonacci (simple examples only) *Real connect: Control flow is the brain of every program — from traffic signal logic to drone autopilot decision-making.*	
3.	C	**Advanced Programming Concepts (C):** 1. 1D array — declaration, input, output, traversal 2. Simple operations — find max, min, sum, average 3. Strings in C — gets/puts, strlen, strcpy, strcmp 4. Pointer — concept of address and pointer variable (concept only, not deep) 5. Structures, 6. File Handling *Real connect: Arrays store sensor readings, image pixels, and stock prices. Pointers are the foundation of memory management in embedded systems.*	20%
4.	Python	**Python Introduction:** 1. Why Python — where it is used (AI, Data Science, Web, Automation) 2. Python basics — variables, data types, print, input, operators 3. Python control flow — if/else, for loop, while loop 4. Python functions — def, return, default arguments *Real connect: Python is #1 in AI/ML, Data Science, and automation — same syntax, more power, less boilerplate.*	20%
5.	Python	**Python Data Structures and Emerging Applications** 1. Python list, tuple, dictionary, set — basic operations only 2. File handling — read and write a text file; read a CSV (2–3 lines) 3. Introduction to NumPy — array, mean, simple arithmetic (5 lines) 4. Introduction to Matplotlib — plot a simple line graph (5 lines) 5. One mini-demo — IoT sensor simulation	20%

CSE Stream

Essentials of Data Structure

Unit	Language	Topics	Weightage
1.	C / Python	**Introduction and Arrays** 1. Data structure — definition, need, classification (linear / non-linear, static / dynamic) 2. Algorithm efficiency — Big O notation: O(1), O(n), O(log n), O(n^2) — concept only 3. Array — insert, delete, search, traverse in C/Python 4. 2D array — matrix addition and transpose *Real connect:NumPy array (AI/ML), image as 2D matrix, sensor data log as array, spreadsheet rows.*	20%

2.	C/Python	**Linked List** 1. Node — data + pointer; why linked list over array 2. Singly linked list — insert (head, tail, position), delete, traverse 3. Doubly linked list — concept and structure only 4. Circular linked list — concept only *Real connect: Browser back/forward = doubly linked list. Music playlist = circular. Python deque = doubly linked list (used in BFS, LRU cache).*	20%
3.	C/Python	**Stack and Queue** 1. Stack — LIFO; push, pop, peek; implementation using array in C/Python 2. Application: balanced parentheses check; function call stack (concept) 3. Queue — FIFO; enqueue, dequeue; implementation using array in C/ Python 4. Application: print spooler; task scheduling queue, Polish Notation *Real connect: Ctrl+Z = Stack. WhatsApp delivery = Queue. Python error traceback IS a call stack. CPU scheduling = queue.*	20%
4.	C/Python	**Searching and Sorting** 1. Linear search, binary search, comparison 2. Bubble sort and insertion sort, Quick Sort and Selection Sort — trace + C/ Python 3. Merge sort — concept and trace only 4. Python built-in sort() and sorted() *Real connect: Binary search in database index. Python sort() uses Timsort. E-commerce ranking, leaderboard, ML data preprocessing = sorting.*	20%
5.	Trace + Python demo	**Trees and Graphs (Concept Level)** 1. Binary tree — terminology (root, leaf, height, level); properties 2. Binary Search Tree — insert, search; inorder/preorder traversal trace 3. Graph — vertex, edge, directed/undirected; adjacency list representation, insertion	20%

CSE Stream

Unit	Description	Weightage (%)
1	**Foundations of the Indian Knowledge System** Definition, scope, characteristics and significance of IKS; the IKS vision of NEP 2020; the fourteen (Chaturdasha) Vidyasthanas — four Vedas, four Upavedas and six Vedangas; Schools of Indian Philosophy: Orthodox and Heterodox Schools; Oral and Written Knowledge Systems, Knowledge validation (pramanas). Case study: relevance of IKS for a present-day engineer.	20
2	**Indian Mathematics, Astronomy and Scientific Thinking** **Number Systems**: Decimal place-value system, zero and infinity; Bhutasankhya and Katapayadi numeration; Pingala's Chandahshastra and binary representation; Sulbasutras; contributions of	20

	Aryabhata, Brahmagupta, Bhaskara and the Kerala School; Indian astronomy and the Panchanga. **Numerical:** encoding numbers using the Katapayadi scheme/Observing Panchang and Reflecting on the Patterns	
3	**Engineering, Technology and Material Sciences** Metallurgy — Wootz steel, zinc distillation and the Delhi Iron Pillar; town planning and architecture (Harappan, Vastu and Shilpa Shastra); water management — stepwells, tanks and irrigation, textile engineering Case study: materials-science analysis of the Iron Pillar/ Architecture of Stupas	20
4	**Health, Wellness, Environment and Governance** Ayurveda (Charaka and Sushruta), Tridosha; Yoga (Patanjali's Ashtanga) and wellness; environment and sustainability in Indian thought; governance and management — Kautilya's Arthashastra; ethics from the Bhagavad Gita. Case study: Yoga for well-being / Arthashastra management principles/ Learnings from Geeta	20
5	**Contemporary Relevance, AI and Applications of IKS** Digital preservation (CSIR-TKDL, IKS Division) and IPR; Sanskrit and computational linguistics — Panini's Ashtadhyayi, NLP and AI for ancient texts; IKS-inspired sustainable engineering and innovation. Case study: AI/NLP applied to a Sanskrit corpus.	20

Teaching-Learning Methodology	Delivery through interactive lectures, documentaries and IKS Division resources, expert talks, heritage field visits, assignments and a mini-project connecting IKS to a contemporary problem, supported by NPTEL / SWAYAM MOOCs, in line with NEP 2020 and OBE.

SE Stream

Professional Communication & Technical Writing

Unit	Description	Weightage (%)
1	**Professional Communication and Workplace Readiness** • Fundamentals of communication: definition, process, importance of communication barriers to communication, 7 C's of communication • Business Correspondence : Emails, Notices, Agenda, Minutes of Meeting, Technical Report and proposal • Résumé, CV and Cover Letter Writing	20

	• Interview Skills, Professional Networking and branding e,g.,LinkedIn , X, ATS Score • Digital Communication Etiquette : online presence and virtual meetings **Assignments/Activities** • Prepare a professional résumé and cover letter. • Draft business emails and meeting minutes. • Participate in a mock interview.	
2	**Grammar and Vocabulary for Technical Writing** • Parts of Speech and Sentence Structure • Common Grammatical Errors, Subject–Verb Agreement, Tenses, and Articles • Technical Vocabulary, Word Formation, and Precision in Language • Tone, Style, and Clarity in Technical English • Editing and Proofreading Techniques for Technical Writing • Grammar and Vocabulary Enhancement through Practical Writing Exercises **Assignments/Activities** • Edit and proofread a technical paragraph to improve grammar, vocabulary, precision, and clarity.	20
3	**Public – Speaking, Presentation and AI-Assisted Communication** • Public speaking technique: Modes, verbal, and non-verbal communication • Understating non-verbal Communication through the *Natyashastra*: *Angika Abhinaya* (Body Language: Posture, Gestures, Facial Expressions), *Vachika Abhinaya* (Voice Modulation and Speech Delivery) , *Sattvika Abhinaya* (Expression of Emotions and Authentic Communication) • Presentation Planning and Delivery • Data Visualization and Slide Design • Elevator Pitch • AI Tools for Presentation Design and Speech Practice, prompt writing for effective communication • Ethical Use of AI in Communication **Assignments/Activities** • Deliver an individual presentation. • Prepare an AI-assisted presentation with proper citations. • Record and evaluate an elevator pitch.	20

CSE Stream

4	**Emotional Intelligence and Critical Thinking for Workplace Communication** • Emotional Intelligence through the *Bhagavad Gita* :Self-awareness ,Self-regulation ,Empathy, Emotional Balance • Critical Thinking through Indian Values and Logical Traditions: *Nyaya* (Logical Reasoning), *Viveka* (Discrimination and Judgment), • Value-based Decision Making • Ethical Problem Solving • Reflective Thinking in Professional Contexts **Assignments/Activities** • Case study analysis using value-based decision making. • Reflective journal on emotional intelligence. • Group discussion on workplace ethical dilemmas.	20
5	**Leadership, Workplace Well-being, and Mindful Communication** • Leadership definition and types • Leadership Communication: Active Listening, Constructive Feedback, Teamwork, and Collaboration • Mental Health, Stress Management, and Work-Life Balance • Effective Communication through Yogic Practices: **Yama** as Principles of Ethical Communication **Ahimsa** – Respectful and Non-violent Communication, **Satya** – Truthfulness and Transparency, **Asteya** – Integrity and Honesty, **Brahmacharya** – Responsible and Balanced Communication, **Aparigraha** – Freedom from Ego, Bias, and Possessiveness. **Niyama** for Personal and Professional Growth, **Shaucha** – Clarity of Thought and Communication, **Santosha** – Contentment and Positive Workplace Relationships, **Tapas** – Self-discipline and Professional Commitment, **Svadhyaya** – Self-reflection and Continuous Learning ,**Ishvara Pranidhana** – Purpose, Humility, and Service-oriented Leadership • **Storytelling through Panchatantra:** *The Lion and the Rabbit* (Leadership and Strategic Communication), *The Merchant's Iron Balance* (Emotional Intelligence and Strategic Conflict Resolution), *The Two Fish and the Frog* (Leadership,	20

CE Stream

	Vigilance and Decision Making), and *The Merchant and the King's Servant* (Respectful Leadership, Workplace Ethics and Employee Relations).	

Introduction to AI & Prompt Engineering

Unit	Topics	Weightage
1.	**Introduction to Artificial Intelligence** 1. Definition and scope of AI; brief history — from Alan Turing to ChatGPT	20%

	1. Types of AI — Narrow AI, General AI, Super AI (concept) 2. AI vs Machine Learning vs Deep Learning — what is the difference 3. How machines learn — data, patterns, and predictions (concept, no maths) 4. AI today — where it already exists in daily life (Google Maps, Spotify, Netflix, Google Photos, voice assistants) 5. Machine Learning Concepts, Learning Strategies- Supervised, Unsupervised, Reinforcement Learning 6. Data, Data Visualization, Data Pre-processing **Class Activity: Students list 5 AI tools they use every day and discuss how each one 'learns'.** *Branch connect: CE — AI in structural health monitoring; ME — AI in predictive maintenance; EE — smart grid; CSE — AI in compilers; BT — drug discovery; AG — crop yield prediction.*	
2.	**Core AI Concepts** 1. Machine learning: Classification Techniques- K-Nearest Neighbors (KNN), Naïve Bayes, Clustering Technique-K-Means 2. Classification metrics-accuracy, precision, recall, F1 score, MAE. 3. Neural networks — concept of neurons, layers, weights (simple diagram) 4. Deep learning — what makes it 'deep'; why it needs GPUs, CNN 5. Computer vision — how AI sees images (face recognition, medical imaging) 6. Natural Language Processing (NLP) — how AI understands text and speech **Class Activity: Show Google Teachable Machine (teachablemachine.withgoogle.com) — train a simple image classifier in 5 minutes, no code.** *Branch connect: ECE — image sensors + AI; BT — protein structure prediction (AlphaFold); AG — plant disease detection from photos; TX — fabric defect detection.*	20%
3.	**Generative AI and Large Language Models** 1. What is Generative AI — AI that creates text, images, code, audio, video	20%

CSE Stream

Unit	Description	Weightage
1.	**Engineering Mechanics and Strength of Materials (Basics):** Types of forces and their effects; concept of stress and strain; beams — types of loads (point load, UDL), engineering materials — metals, polymers, ceramics, composites (properties and uses). Applications: frame of a laptop/server rack, load on a tower antenna, material selection for a robot arm, chassis of an electric vehicle.	20%
2.	**Thermodynamics and Heat Transfer Basics:** System and surroundings; laws of thermodynamics; heat transfer — conduction, convection, radiation. Applications: cooling fan and heat sink in a computer processor, AC in a server room/data centre, EV battery thermal management.	20%

3.	**Machine Elements and Power Transmission:** Fasteners — bolts, screws, keys (types and use); shafts and couplings — rigid and flexible; bearings — ball bearing and roller bearing; belts and chains. Applications: gear train in a robotic joint, belt drive in a 3D printer, ball bearings in drone motor.	20%
4.	**Manufacturing Processes:** Casting, forming, machining (drilling, milling, grinding), joining — soldering, brazing; additive manufacturing — 3D printing, CNC machining concept. Applications: 3D-printed drone frame, cast aluminium heat sink, welded server rack.	20%
5.	**Mechatronics, Robotics, and Emerging Mechanical Systems:** Introduction to mechatronics; actuators — DC motor, stepper motor, servo motor, hydraulic and pneumatic actuators, introduction to robotics (Cartesian, SCARA, articulated), drones — airframe, propulsion, flight controller concept and Industry 4.0.	20%

Course Content — Design Thinking Modules (AICTE Aligned)

Module	Description	Hours
1.	**Module 1 – Introduction to Design Thinking & Innovation:** What is Design Thinking; difference between design thinking and conventional problem solving; the five stages (Empathize, Define, Ideate,	6 hrs

	Prototype, Test); innovation vs invention; examples of design thinking success stories (Apple, Airbnb, IDEO, Indian startups); creativity and the innovation mindset; introduction to the team project.	
2.	**Module 2 – Empathize: Understanding the User:** Human-centred design; observation and user research techniques; conducting interviews and surveys; empathy mapping; identifying user needs, pain points, and latent needs; stakeholder analysis; developing personas. Studio Activity: field observation and user interviews for the chosen problem.	8 hrs
3.	**Module 3 – Define & Ideate:** Synthesising research; framing the problem statement (Point of View); 'How Might We' questions; brainstorming rules and techniques; SCAMPER, mind mapping, lateral thinking; idea selection and prioritisation (impact-effort matrix, dot voting); concept development. Studio Activity: define problem statement and generate/short-list solution ideas.	10 hrs
4.	**Module 4 – Prototype:** Purpose of prototyping; types of prototypes (paper, low-fidelity, working model, digital mock-up); rapid prototyping with low-cost materials; storyboarding; introduction to prototyping tools (cardboard, clay, 3D printing basics, Tinkercad, Figma/Canva for digital prototypes). Studio Activity: build a low-cost prototype of the team's solution.	12 hrs
5.	**Module 5 – Test, Iterate & Pitch:** User testing and feedback collection; iteration and refinement based on feedback; failure as learning; introduction to business model canvas (basic); value proposition; storytelling and pitching; preparing and delivering the final project pitch. Studio Activity: test prototype with users, iterate, and present the final pitch.	9 hrs

CSE Stream

Course Outcomes (COs)

CO	Course Outcome Statement	KL

Define Stage	Problem statement (POV) + How-Might-We	8
Ideate Stage	Brainstorm record + short-listed concept	8
Prototype Stage	Low-cost working prototype / model	12
Design Journal	Team logbook / portfolio of the whole process	7
Team Participation	Individual contribution + collaboration	5

Suggested Project Themes (Teams may choose or propose their own)

1. Campus & Community — waste management, water conservation, accessibility for the differently-abled, canteen queue management
2. Agriculture & Rural — low-cost farming tools, water-saving irrigation, crop storage, farmer information access
3. Health & Wellbeing — elderly care aids, hygiene solutions, mental wellness, first-aid tools
4. Sustainability & Environment — plastic reduction, energy saving, green mobility, air-quality awareness
5. Education & Skilling — learning aids, tools for rural schools, assistive learning for disabilities
6. Smart Living — affordable home automation, safety devices, smart utilities

Suggested Learning Resources

1. AICTE Model Curriculum – Design Thinking (aicte-india.org)
2. The Design of Everyday Things – Don Norman
3. Change by Design – Tim Brown (IDEO)
4. Design Thinking Bootleg – Stanford d.school (dschool.stanford.edu, free)
5. IBM Enterprise Design Thinking – ibm.com/design/thinking (free)
6. NPTEL – Design Thinking: A Primer (IIT Madras)

CSE Stream

Unit	Description	Hours
1.	**Environment and Ecosystems:** Environment — definition, components (atmosphere, hydrosphere, lithosphere, biosphere); ecosystem — structure and function; food chain, food web, energy flow (basic concept); biodiversity — meaning, importance, and threats; need for conservation. Awareness focus: local ecosystems and why they matter.	6 hrs
2.	**Natural Resources and their Conservation:** Renewable and non-renewable resources; water resources — scarcity and conservation; forest and land resources; energy resources (brief); over-exploitation of resources; the concept of the 3 R's — Reduce, Reuse, Recycle; rainwater harvesting (concept). Awareness focus: simple resource-saving practices in daily life.	6 hrs
3.	**Environmental Pollution:** Air pollution — causes, effects, control (basic); water pollution — sources, effects, prevention; soil pollution; noise pollution; solid waste management — segregation, composting, recycling; e-waste and plastic waste (awareness). Awareness focus:	6 hrs

	pollution in the student's own city/campus and how to reduce it.	
4.	**Climate Change and Sustainable Development:** Global warming and greenhouse effect (simple explanation); climate change — causes and impacts; carbon footprint (concept + how to reduce); sustainable development — definition and need; UN Sustainable Development Goals (SDGs) — overview of the 17 goals; renewable energy — solar, wind, biomass (awareness level). Awareness focus: calculating and reducing one's own carbon footprint.	6 hrs
5.	**Environmental Protection, Law and the Role of Engineers:** Environmental protection measures; brief overview of environmental laws in India (Environment Protection Act, Water Act, Air Act — awareness only, no legal detail); role of individuals, society, and technology in environmental protection; green campus and green buildings (concept); role of engineers in sustainable design and clean technology; environmental ethics. Awareness focus: how each engineering branch can contribute to sustainability.	6 hrs

Suggested Field / Awareness Activities (any one to be completed)

1. Visit a nearby lake, river, park, or forest and prepare a short observation report
2. Conduct a waste segregation / plastic-free drive on campus and document it
3. Prepare a poster or short video on an environmental issue (water, air, plastic, climate)
4. Calculate your household or personal carbon footprint and suggest ways to reduce it
5. Plant and maintain a sapling; document its growth over the semester (Green initiative)
6. Survey energy or water use in a hostel/department and suggest saving measures

CSE Stream